Safety Data Sheet

Creation Date: 2016/07/01	**SDS NO.: 2071E**
Product Name: tetrahydrofuran	**Version: V1.0.0.3**

SECTION 1 Identification of the chemical and supplier

Chemical Name: tetrahydrofuran

Synonyms: diethylene oxide
CAS No.: 109-99-9
EC No.: 203-726-8
Molecular Formula: C4H8O
Recommended Uses: Please consult manufacturer.
Restrictions on Use: Please consult manufacturer.
Name of the company: Weifang Zhengxing Risheng Chemical Co., Ltd.

Address of the company: Tuanjie Rd.25th, Changle Town.Weifang City, Shandong Province, 262404,China
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SECTION 2 Hazards identification

| Emergency overview

Liquid. Highly flammable, its vapor and air mixture can form explosive mixture. Serious irritating to eyes. Irritating to respiratory system. slight risk of cancer.

| Hazard classification according to GHS

According to series standards of GB 30000-2013: Rules for classification and labelling of chemicals. (Please refer to 16th chapter of SDS), hazard classification as following: Flammable Liquids, Category 2; Serious Eye Damage/Irritation, Category 2A; Specific Target Organ Toxicity-Single Exposure : Criteria for respiratory tract irritation, Category 3; Carcinogenicit, Category 2.

| Label elements

Hazard pictograms

Signal word: Danger

Hazard statements: Highly flammable liquid and vapour, Causes serious eye irritation, May cause respiratory irritation, Suspected of causing cancer.

Precautionary statements

Prevention: Obtain special instructions before use.Do not handle until all safety precautions have been read and understood.Keep away from heat, hot surfaces, sparks,open flames and other ignition sources. Nosmoking.Keep container tightly closed.Ground and bond container and receiving equipment.Use non-sparking tools.Take action to prevent static discharges.Avoid breathing dust/fume/gas/mist/vapours/spray.Wash ... thoroughly after handling.Use only outdoors or in a well-ventilated area.Wear protective gloves/protective clothing/eye protection/face protection

Response: Call a POISON CENTER/doctor,if you feel unwell.IF INHALED: Remove person to fresh air and keep comfortable for breathing.IF exposed or concerned: Get medical advice/ attention.If eye irritation persists: Get medical advice/attention. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Storage: Store locked up.Store in a well-ventilated place. Keep container tightly closed.Store in a well-ventilated place. Keep cool.

Disposal: Dispose of contents/container in accordance with local/regional/national/ international regulations.

| Hazard description

Physical and chemical hazards

Highly flammable liquids, its vapor and air mixture can form explosive mixture.

Health hazards

Inhalation of vapours, especially for prolonged periods, may produce respiratory irritation and occasionally, distress. Inhalation of the product may produce adverse health effects or irritation of the respiratory tract following discomfort. Accidental ingestion of the product may be harmful to the health of the individual. Entry into the blood-stream, through, for example, cuts, abrasions or lesions, may produce systemic injury with harmful effects. This product may cause serious eye irritation. Severe inflammation may be expected with pain following direct contact with the eye. This product may cause temporary discomfort following direct contact with the eye.

Environmental hazards

Please refer to 12th chapter of SDS.

SECTION 3 Composition/information on ingredients

√Substance Mixture

Component	Concentration/Range	CAS No.
Tetrahydrofuran	>= 99.0	109-99-9

SECTION 4 First aid measures

| Description of first aid measures

General advice: Immediate medical attention is required. Show this safety data sheet (SDS) to the doctor in attendance.

Skin contact: Take off contaminated clothing and shoes immediately. Wash off with plenty of water for at least 15 minutes and consult a physician if fell uncomfortable.

Eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician if fell uncomfortable.

Inhalation: Move victim into fresh air. If breathing is difficult, give oxygen. Do not use mouth to mouth resuscitation if victim ingested or inhaled the substance. If not breathing, give artificial respiration and consult a physician immediately.

Ingestion: Do not induce vomiting. Never give anything by mouth to an unconscious person. Call a physician or Poison Control Center immediately.

Advice for protecting the rescuer: Remove all sources of ignition and increase ventilation. Avoid contact with skin and eyes. Avoid inhalation of vapor or mist. Use personal protective equipment including respirator .

Special note to the doctor: Treat symptomatically. Symptoms may be delayed.

SECTION 5 Firefighting measures

| Hazard characteristics

Will form explosive mixtures with air. Fire exposed containers may vent contents through pressure relief valves thereby increasing fire intensity and/ or vapour concentration. Vapours may travel to source of ignition and flash back. Liquid and vapour are flammable. Containers may explode when heated. Fire exposed containers may vent contents through pressure relief valves . May expansion or decompose explosively when heated or involved in fire.

| Extinguishing method and media

Suitable extinguishing media: dry chemical, carbon dioxide or alcohol-resistant foam.

Unsuitable extinguishing media: Do not use a solid water stream as it may scatter or spread fire.

| Fire precautions and measures

As in any fire, wear self-contained breathing apparatus (MSHA/NIOSH approved or equivalent)and full protective gear. Fight fire from a safe distance, with adequate cover. Prevent fire extinguishing water from contaminating surface water or the ground water system.

SECTION 6 Accidental release measures

| Personal precautions, protective equipment and emergency procedures

Avoid breathing vapors and contacting with skin and eye. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas. Emergency personnel wear positive pressure self-contained breathing apparatus. Wear protective and anti-static clothing. Wear chemical impermeable gloves. Ensure adequate ventilation. Remove all sources of ignition. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Use personal protective equipment. Avoid breathing vapours, mist, gas or dust.

| Environmental precautions

Prevent further leakage or spillage if safe to do so. Discharge into the environment must be avoided.

| Methods and materials for containment and cleaning up

Absorb spilled material in dry sand or inert absorbent. In case of large amount of spillage, contain a spill by bunding. Adhered or collected material should be promptly disposed of, in accordance with appropriate laws and regulations. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

SECTION 7 Handling and storage

| Precautions for handling

Avoid inhalation of vapors. Use only non-sparking tools. To prevent fire caused by electrostatic discharge steam, equipment on all metal parts should be grounded. Use explosion proof equipment. Handling is performed in a well ventilated place. Wear suitable protective equipment. Avoid contact with skin and eyes. Keep away from heat/sparks/open flames/ hot surfaces. Take precautionary measures against static discharges.

| Precautions for storage

Keep containers tightly closed . Keep containers in a dry, cool and well-ventilated place. Keep away from heat/sparks/open flames/ hot surfaces. Store away from incompatible materials and foodstuff containers.

SECTION 8 Exposure controls/personal protection

| Control parameters

Occupational Exposure limit values

COMPONENT	STANDARD	TYPE	STANDARD VALUE	REMARK
Tetrahydrofuran	GBZ 2.1-2007	PC-TWA	300 mg/m3	-
		PC-STEL	-	

Biological limit values

No information available.

Monitoring methods

EN 14042 Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. GBZ/T 160.1~GBZ/T 160.81-2004 Determination of toxic substances in workplace air (Series standard) .

| Engineering controls

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Set up emergency exit and necessary risk-elimination area.

| Respiratory protection

If exposure limits are exceeded or if irritation or other symptoms are experienced, use a full-face respirator with multi-purpose combination (US) or type AXBEK (EN 14387) respirator cartridges.

| Eye protection

Tightly fitting safety goggles (approved by EN 166(EU) or NIOSH (US).

| Skin and body protection

Wear fire/flame resistant/retardant clothing and antistatic boots.

| Hand protection

Wear protective gloves (such as butyl rubber) , passing the tests according to EN 374(EU),US F739 or AS/NZS 2161.1 standard.

| Other protection

Smoking, eating and drinking water is forbidden in the workplace. After work, shower and change clothes. To maintain good health habits.

SECTION 9 Physical and chemical properties

Appearance: colorless transparent liquid	
pH (Specified the concentration) : 5	**Odor:** No information available
Initial boiling point and boiling range(℃): 66	**Melting point/freezing point(℃):** -108.5
Vapor density(Air = 1): 2.5	**Density:** No information available
Saturated vapor pressure (kPa): 19.3	**Relative density(Water=1):** 0.891
Evaporation rate: No information available	**Viscosity(mm2/s):** No information available
Flash point(℃): -14.5	**N-octanol/water partition coefficient:** No information available
Decomposition temperature(℃): No information available	**Auto-ignition temperature(℃):** 321
Upper/lower explosive limits[%(v/v)]: Upper limit : 11.8 ; Lower limit : 2	
Solubility: Miscible with water	**Flammability:** Not applicable

SECTION 10 Stability and reactivity

| Stability

Stable under proper operation and storage conditions.

| Incompatible materials

Strong acids, strong alkalis, strong oxidants and strong reducing agents.

| 应 Conditions to avoid

Incompatible materials, heat, flame and spark.

| Hazardous reactions

May react with strong acids, strong alkalis, strong oxidants and strong reducing agents.

| Decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11 Toxicological information

Acute toxicity

COMPONENT	CAS NO.	LD_{50}(ORAL)	LD_{50}(DERMAL)	LC_{50}(INHALATION)-4H
Tetrahydrofuran	109-99-9	1650mg/kg(Rat)	No information available	No information available

Carcinogenicity

ID	CAS NO.	COMPONENT	IARC	NTP
1	109-99-9	Tetrahydrofuran	Not Listed	Not Listed

Skin irritation/corrosion

No information available

Eye irritation/corrosion

Causes serious eye irritation(Category 2A)

Skin sensitization

No information available

Respiratory sensitization

No information available

Germ cell mutagenicity

No information available

Reproductive toxicity

No information available

STOT-single exposure

May cause respiratory irritation(Category 3)

STOT-repeated exposure

No information available

Aspiration hazard

No information available

SECTION 12 Ecological information

Acute aquatic toxicity

COMPONENT	CAS NO.	FISH	CRUSTACEANS	ALGAE
Tetrahydrofuran	109-99-9	LC_{50}: 2160mg/L (96h)(Fish)	No information available	No information available

Chronic aquatic toxicity

No information available.

| Persistence and degradability

No information available

| Bioaccumulative potential

No information available

| Mobility in soil

No information available

| Other deleterious effect

No information available.

SECTION 13 Disposal considerations

| Disposal methods

Product: If medical advice is needed, have product container or label at hand.

Contaminated packaging: Containers may still present chemical hazard when empty. Keep away from hot and ignition source of fire. Return to supplier for recycling if possible.

| Disposal considerations

Refer to section 13.1and 13.2.

SECTION 14 Transport information

| UN number: 2056

| UN proper shipping name: TETRAHYDROFURAN

| Transport hazard class: 3

| Packing group: Ⅱ

| Label

| Marine pollutant (Yes/No): No

| Methods of packing

Metal drum, removable head. Ampoule outside the ordinary wooden box. Threaded glass, metal cover pressure bottles, plastic bottles or metal (cans) outside the ordinary wooden box etc. Packaging as recommended by manufacturer.

| Precautions for transport

Shipment of the goods vehicle exhaust pipe must be equipped with fire retardant devices, prohibit using mechanical equipment and toolsof which easy to produce sparks . Transit should be anti-exposure, anti-rain, anti-high temperature. Transportation used tank (tank) cars should be grounded chain, tank can be installed

to reduce the partition hole static electricity shocks. Strictly prohibited shipping or transportation withoxidants, acids, food and food additives etc. When bulk transport,Prohibit the use of cement or wooden boats. Transport vehicles should be equipped with the appropriate variety and quantity of fire equipment and emergency equipment leakage during transport. Before transport, should be preceded by checking whether container integrity, sealing. The transport unit must be placarded and marked in accordance with relevant transporting requirements.

SECTION 15 Regulatory information

| Regulatory information

COMPONENT	A	B	C	D	E	F	G	H
Tetrahydrofuran	Listed	Not Listed	Not Listed	Not Listed	Not Listed	Not Listed	Not Listed	Not Listed

【A】 Catalog of Hazardous Chemicals(2015 Edition), Notice 5th 2015, China State Administration of Work Safety.

【B】 Catalog of Priority Hazardous Chemicals for Environment Management, Notice 33th 2014, Ministry of Environmental Protection of PRC.

【C】 List of Toxic Chemicals Restricted to be Imported/Exported in China, Notice 85th 2013, Ministry of Environmental Protection of PRC.

【D】 Catalog of Stupefacient and Psychotropic Substances(2013Edition), Notice 230th 2013, China Food and Drug Administration.

【E】 Catalog of Hazardous Chemicals for Priority Management (First and Second batches) , Notice 95th, 2011, Notice 12th 2013, China State Administration of Work Safety.

【F】 List of Ozone Depletion Chemicals Controlled to be Imported/Exported in China (First to Sixth batches) , Notice from 2000 to 2012 Ministry of Environmental Protection of PRC.

【G】 Dangerous Chemicals Directory Used to Manufacure Exploder(2011 Edition), Notice 25th Nov. 2011, Ministry of Public Security of PRC1.

【H】 Catalog of Highly Toxic Chemicals, Notice 142th 2003, China Ministry of Health.

SECTION 16 Others

Latest Revision Date: 2016/07/01

| Revision explanation

This Safety Data Sheet (SDS) was prepared according to standards of《Safety data sheet for chemical products——Content and order of sections》(GB/T 16483-2008) and《Guidance on the compilation of safety data sheet for chemical products》(GB/T 17519-2013) . The result of GHS classification according to《Guidance on the Catalog of Hazardous Chemicals(2015 Edition) (trial)》 and series standards of《Rules for classification and labelling of chemicals》 (GB 30000.2-2013~GB 30000.29-2013).

| Reference

【1】 IPCS: The International Chemical Safety Cards (ICSC) ,website:http://www.ilo.org/dyn/icsc/showcard.home.
【2】 IARC , website:http://www.iarc.fr/.
【3】 OECD: The Global Portal to Information on Chemical Substances, website: http://www.echemportal.org/echemportal/index?pageID=0&request_locale=en.
【4】 CAMEO Chemicals, website:http://cameochemicals.noaa.gov/search/simple.
【5】 NLM: ChemIDplus, website:http://chem.sis.nlm.nih.gov/chemidplus/chemidlite.jsp.

【6】 EPA: Integrated Risk Information System, website:http://cfpub.epa.gov/iris/.
【7】 U.S. Department of Transportation: ERG, website:http://www.phmsa.dot.gov/hazmat/library/erg.
【8】 Germany GESTIS-database on hazard substance, website:http://gestis-en.itrust.de/.

Abbreviations and acronyms

CAS –Chemical Abstracts Service

CMR - Carcinogens, mutagens or substances toxic to reproduction

PC-STEL- Short term exposure limit

PC-TWA - Time Weighted Average

DNEL - Derived No Effect Level

IARC - International Agency for Research on Cancer

RPE - Respiratory Protective Equipment

PNEC –Predicted No Effect Concentration

LC_{50} - Lethal Concentration 50%

LD_{50} - Lethal Dose 50%

NOEC -No Observed Effect Concentration

EC_{50} - Effective Concentration 50%

PBT - Persistent, Bioaccumulative, Toxic

POW - Partition coefficient Octanol:Water

BCF - Bioconcentration factor (BCF)

vPvB - very Persistent, very Bioaccumulative

IMDG-International Maritime Dangerous Goods

ICAO/IATA-International Civil Aviation Organization/International Air Transportation Association

UN-The United Nations

ACGIH-American Conference of Governmental Industrial Hygienists

NFPA-National Fire Protection Association

OECD-Organization for Economic Co-operation and Development

Disclaimer

This Safety Data Sheet (SDS) was prepared according to GB/T16483 and GB/T17519. The data included was derived from international authoritative database and provided by the enterprise. Other information was based on the present state of our knowledge. We try to ensure the correctness of all information. However, due to the diversity of information sources and the limitations of our knowledge, this document is only for user's reference. Users should make their independent judgment of suitability of this information for their particular purposes. We do not assume responsibility for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of the product.